

Executive Hotel Booking EDA

Exploratory Data Analysis of Hotel Booking Data

1. Introduction

This report presents an exploratory analysis of hotel booking data. The analysis focuses on data quality, booking and cancellation patterns, pricing, market segments, and relationships between important booking variables.

2. Dataset Overview

The dataset contains hotel reservation records with information about hotel type, booking status, lead time, arrival details, length of stay, guests, market segment, distribution channel, deposit type, average daily rate (ADR), previous booking history, and special requests.

3. Data Quality and Preprocessing

- The dataset structure, dimensions, variables, and data types were reviewed.
- Missing values and inconsistent categorical values were assessed.
- Exact duplicate records were identified and removed.
- 180 duplicate records were removed, leaving 25,000 unique records.
- Numerical variables were checked for unusual observations and outliers before analysis.
- Variables were cleaned and converted where necessary for analysis.

4. Exploratory Data Analysis

Descriptive statistics, univariate analysis, bivariate analysis, group-wise analysis, and correlation analysis were performed. Matplotlib and Seaborn were used for visualization.

Cancellation Analysis

- The overall cancellation rate is approximately 72.2%.
- Bookings made 181–365 days in advance have an 85.8% cancellation rate.
- Bookings made within 7 days have a 43.2% cancellation rate.
- Non Refund bookings have a 93.0% cancellation rate.

Hotel Type and Pricing

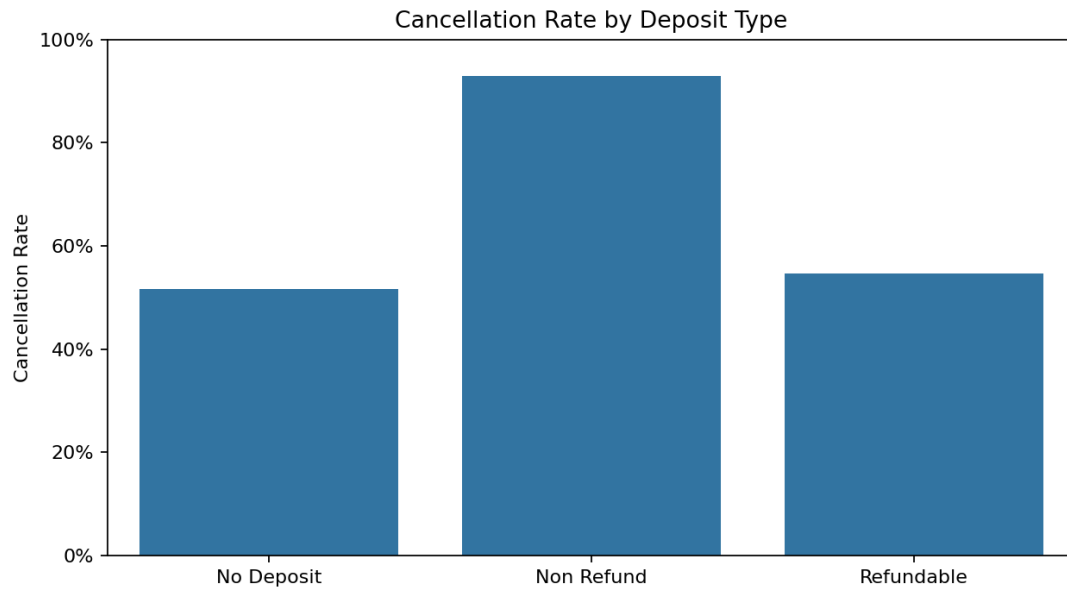
- Resort Hotels have a higher average daily rate than City Hotels.
- Average ADR is approximately 173.70 for Resort Hotels and 120.72 for City Hotels.

Market Segment

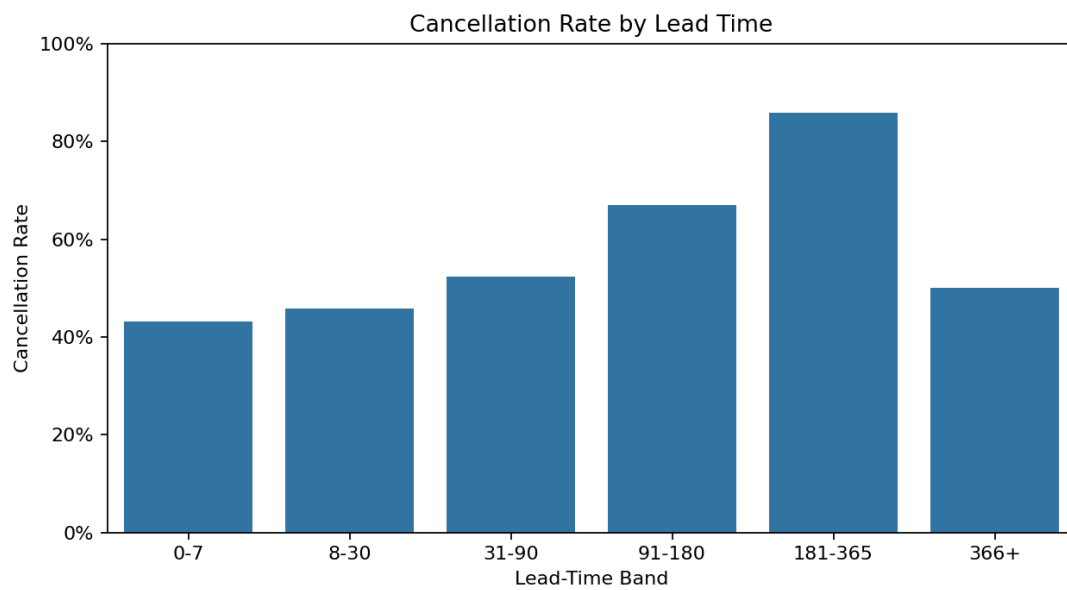
- Online Travel Agencies represent a major source of bookings.
- The Online TA segment has a cancellation rate of approximately 75.1%.

5. Visual Analysis

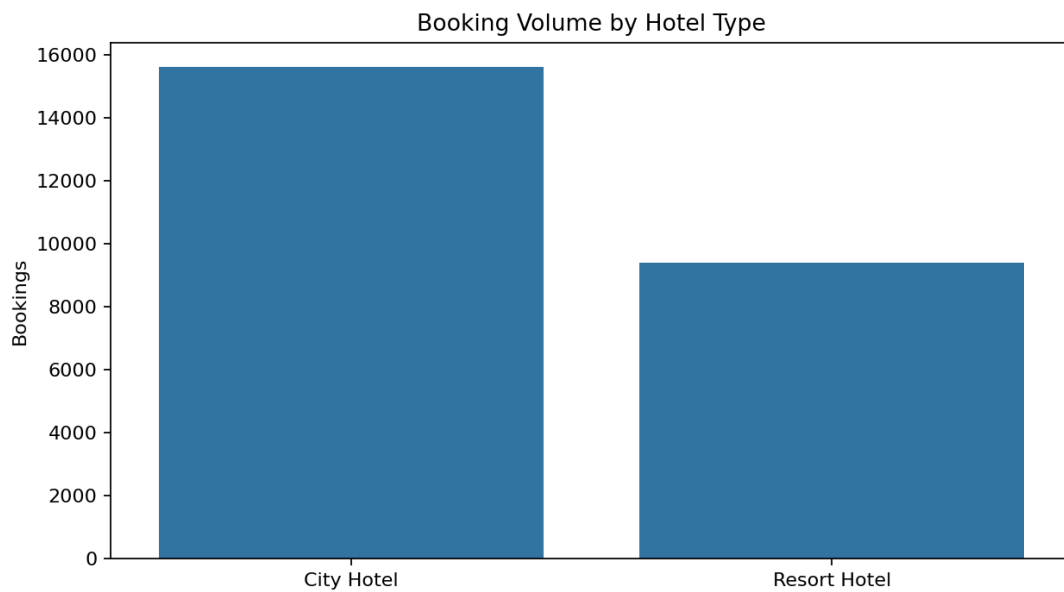
Cancellation by Deposit Type



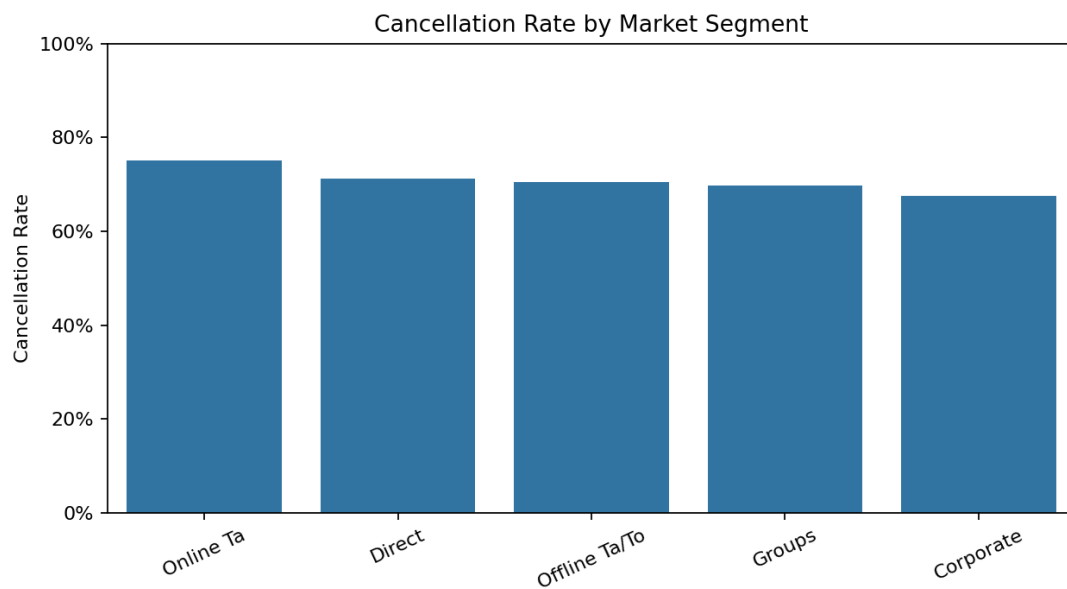
Cancellation by Lead-Time Group



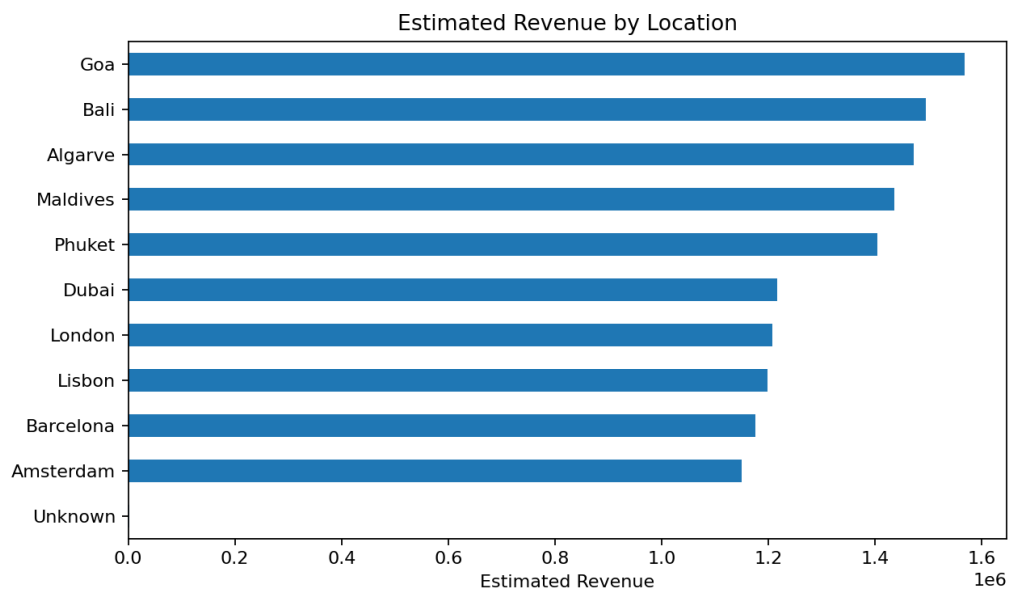
Cancellation by Hotel Type



Cancellation by Market Segment



Revenue by Location



6. Key Business Insights

High cancellation rate

A large proportion of bookings are cancelled, making cancellation management an important business issue.

Longer lead times have higher cancellation risk

Bookings made further in advance show substantially higher cancellation rates.

Deposit type is strongly associated with cancellations

Non Refund bookings show particularly high cancellation levels.

Online TA has a volume-versus-cancellation trade-off

Online Travel Agencies contribute substantial booking volume but also have a high cancellation rate.

Hotel types have different pricing patterns

Resort Hotels have a higher ADR than City Hotels.

7. Recommendations

- 1. Use additional confirmation and reminder measures for long-lead bookings.
- 2. Review deposit and payment policies for booking groups with higher cancellation risk.
- 3. Evaluate Online Travel Agency performance using cancellation-adjusted revenue and realized bookings.
- 4. Use differentiated pricing strategies for Resort and City Hotels.
- 5. Include lead time in cancellation and occupancy forecasting.
- 6. Monitor booking-channel and cancellation performance regularly.
- 7. Consider cancellation, ADR, occupancy, and channel metrics together when making revenue decisions.

8. Conclusion

The analysis shows meaningful differences in booking outcomes based on lead time, deposit type, market segment, and hotel type. These patterns provide useful information for improving cancellation management, pricing decisions, channel evaluation, and operational planning.